# TfL "What-If" Simulation Requirements Document

## 1. Introduction

### 1.1 Purpose

This document defines the functional and non-functional requirements for the London Underground Simulation Project. The system will simulate the London Underground network using real Transport for London (TfL) data to allow users to visualise routes, analyse disruptions and test network resilience through interactive scenario planning.

### 1.2 Scope

The system will provide an interactive, web-based simulation of the London Underground network. It will integrate real-time service data, support shortest-path routing, visualising network activity and allowing manual disruption testing.

## 2. System Overview

The system will consume TfL API data to construct a graph representation of the Underground network. A pathfinding algorithm will calculate optimal routes while the visual interface will display the network and simulation agents. Users may observe live conditions or create hypothetical disruption scenarios using the "What-If" simulation feature.

## 3. Functional Requirements

### FR-01 Network Data Integration

The system shall retrieve station and stop hierarchy data from the TfL API.

### FR-02 Station Hierarchy Mapping

The system shall represent stations as parent hubs with child nodes for platforms and entrances.

### FR-03 Graph Construction

The system shall construct a connected graph representing all stations and lines.

### FR-04 Route Calculation

The system shall calculate the shortest route between two selected stations.

### FR-05 Travel Time Display

The system shall display estimated travel times for calculated routes.

### FR-06 Real-Time Status Updates

The system shall retrieve live line status information and update the network accordingly.

### FR-07 Dynamically Re-Routing

The system shall automatically recalculate routes when disruptions occur.

### FR-08 Interactive Map

The system shall display the Underground network on an interactive map.

### FR-09 Agent Simulation

The system shall simulate trains or passenger flows moving along network routes.

### FR-010 Scenario Planning

The system shall allow users to manually close or reopen stations and lines.

### FR-011 Visual Disruption Indicators

The system shall visually highlight closed or disrupted stations and lines.

### FR-012 Crowd Simulation

The system may simulate overcrowding at stations based on capacity data.

## 4. Non-Functional Requirements

### 4.1 Performance

- System shall calculate routes within 2 seconds.
- Map interactions shall remain responsive under normal usage.

### 4.2 Reliability

- The system shall continue operating if the TfL API is temporarily unavailable using cached data.

### 4.3 Usability

- Interface shall be intuitive for first-time users.
- All controls shall be clearly labelled.

### 4.4 Scalability

- System shall support additional transport lines in the future.

### 4.5 Maintainability

- Codebase shall follow modular architecture principles.

### 4.6 Portability.

- The system shall run on modern web browsers and phone browsers intuitively.